Fiscal Year:	FY 2025	**Task Last Updated:**	FY 05/02/2025
PI Name:	Froio-Blumsack, Danielle M.S.		
Project Title:	Stabilized Foods for Use in Extended Spaceflight: Preservation of Shelf-Life, Nutrient Content and Acceptability		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	01760	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Crew Health NNJ11ZSA002NA
Start Date:	04/21/2014	**End Date:**	05/31/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Final report date changed to 05/31/2024 per T. Sirmons/HRP JSC (Ed., 4/29/24) NOTE: End date changed to 12/31/2023 per T. Sirmons/JSC (Ed., 9/12/23) NOTE: End date changed to 09/30/2023 per T. Sirmons/JSC (Ed., 3/2/23) NOTE: End date changed to 10/31/2022 per C. Ribeiro/HHC element/JSC HRP (Ed., 11/18/21)		
Key Personnel Changes/Previous PI:	Kerry Candlen (ORISE Contractor / UML PhD Candidate - Plastics Engineering) joined the research team in 2022		
COI Name (Institution):			
Grant/Contract No.:	NNJ13HA911 IAA		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	ED. NOTE (June 2021): Principal Investigator (PI) Danielle Froio-Blumsack continues work initiated with PI Dr. Ann Barrett, whose work covered the first 3 years of this contract with U.S. Army. The objective of this effort is to develop shelf stable, highly acceptable food with increased nutrient (vitamins) stability for extended space missions utilizing innovative processing and packaging technologies. Selected packaging technologies will be investigated with research focused on the interaction of packaging material with various innovative sterilization processes such as microwave heating, irradiation, and high pressure treatment. The availability of highly nutritious and health-promoting food is a factor that is a significant prerequisite for prolonged space travel. The design of feeding and nutritional strategies for multi-year, non-resupplied flights is an undertaking requiring substantial research and development; it is also an endeavor and that could be founded upon our existing, considerable knowledge and experience base at the U.S. Army Combat Capabilities Development Command Soldier Center.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The study will yield strategies for the development of extremely stable, nutrient-dense foods and the development of packaging materials compatible with new quality-preserving sterilization techniques. While this work is specifically important to the health of astronauts, its significance also extends to the research that is critical to the mission of the Natick Soldier Research Development and Engineering Center: to support and promote the nutritional health of the Warfighter on extended missions with little or no means of resupply.
Task Progress:	Samples were stored for 6 years (72 months) at 70°F, and pulled and evaluated at yearly increments. Microbiological testing was conducted at each pull to confirm sustained commercial sterility. Microbiological testing was conducted following AOAC Official Method 997.02 for Yeast and Mold Counts in Foods and AOAC Official Method 990.12 for Aerobic Plate Count, with five pouches from each sample group tested. Sensory analysis was conducted through the year 3 (36-month) pull, at which time samples fell below the threshold scoring value of 5 on a 9-point hedonic scale, and future analysis would not provide meaningful data. It should be noted that quality ratings were lower than anticipated at the start of the study, with beginning quality scoring in the range of 6. It is expected that quality would have sustained over a longer storage time, had the initial quality of the CCC been at a higher level. [Ed. Note: CCC refers to pouches containing a Creamy Cajun Chicken (CCC) formulation fortified with two times the spaceflight requirement of vitamins A, B1, B9, C, and E.] Vitamin analysis was conducted to monitor vitamin stability over time. Vitamins E, B9, B1, A, and C were analyzed pre-processing, time zero (immediately after processing) and then again after 12, 24, 36, 48, and 72 months' storage at 70°F. Vitamin analysis was conducted by Eurofins (formerly Covance Co.) (Madison WI), using American Organization of Analytical Chemists (AOAC) International methods for vitamins A, B1, B9, C, and E. For irradiated samples, Vitamin E was the only vitamin analysis performed at the 48 and 72-month storage points, due to vitamin loss for other vitamins falling well below the spaceflight requirement at this time point. Pouch integrity was described by Young's modulus, seal strength, and burst strength. Young's modulus and tensile strength was analyzed in accordance with ASTM D882, "Standard Test Method for Tensile Properties of Thin Plastic Sheeting" using an Instron®5965 Tensile Testing Machine (Instron, Norwood, MA). Seal integrity was analyzed in accordance with ASTM F88 "Standard Test Method for Seal Strength of Flexible Barrier Materials". Burst strength was conducted following ASTM F2054 "Standard Test Method for Burst Testing of Flexible Package Seals Using Internal Air Pressurization within Restraining Plates". Barrier performance was measured in terms of oxygen transmission rate (OTR) and water vapor transmission rate (WVTR). Oxygen transmission rate (OTR) was conducted using a MOCON Ox-tran 2/21(control, initial, 12, 24, and 36-month data) and Ox-tran 2/22 (10x) (48 and 70-month data) (MOCON, Minneapolis, MN) at 50% RH and 73.4oF, in accordance with ASTM D3985, "Standard Test Method for Oxygen Gas Transmission Rate Through Plastic Film and Sheeting Using a Coulometric Sensor". A MOCON Permatran 3/33 (control, initial, 12, 24, and 36-month data) and MOCON Permatran 3/34 (G) (48-month and planned 70-month data) was used to test water vapor transmission rate (WVTR) at 90% RH and 100°F, in accordance with ASTM F1249, "Standard Test Method for Water Vapor Transmission Rate Through Plastic Films and Sheeting Using a Modulated Infrared Sensor". Rough handling studies were conducted on pouches after they completed the six-year storage study. This allows for evaluation of pouch integrity of the various pouches/processing combinations under simulated distribution conditions, including drop and vibration. The samples were packed out using the method used for military rations, in which one pouch is packed into an assembled MRE (meal ready to eat) and then 12 MREs are packed into a case. The cases are then stored at ambient conditions, high temp (100F), low temp (-20F), for a minimum of 72 hours before being subjected to drop and vibration testing. Items were drop tested at a height of 21". Drops are directed at the manufacture's corner of the box, the edges of the box that connect to the manufacture's corner, and every face of the box. Totaling 10 drops. The Drop Testing is tested in accordance with ASTM-D5276: Drop Test of Loaded Containers by Free Fall. Items were also put onto a vibration table for 1 hour at 268 RPM. Vibration Testing was conducted in accordance with ASTM-D999: Methods for Vibration Testing of Shipping Containers. Upon completion of Rough Handling Procedures (Both Drop and Vibration Testing), all cases undergo visual inspection. Pouches were inspected for failures (leaks) and for damage. Retention of vitamins E and A was relatively unchanged over the four-year storage study for MATS and retorted products. Conversely, all processing methods resulted in product that showed a significant decrease in vitamins B1 and C after six years; however, MATS showed higher concentration of these two vitamins in comparison to other processing methods. A decrease in Young's modulus, seal strength, and burst strength was observed for both pouches A and B, but did not impact overall integrity of the pouch in general or during rough handling. Evaluating pouches for rough handling at the conclusion of the extended storage study revealed that loss of mechanical properties observed during tensile, seal, and burst testing did not impact pouch performance during rough handling at ambient and extreme temperatures. OTR and WVTR fluctuated over the course of the study, and exceeded the requirement in some instances for Pouch A and B. Without sensory data to correlate to barrier performance of the pouches over the course of the study, it is not possible to determine if the increase in OTR and WVTR would impact food quality. Microbial analysis was completed and all samples passed, indicating that the change in barrier performance did not affect food safety. Studies at DEVCOM SC (Natick) have shown that Pouch A (Toppan Pouch GL/RD) meets the three-year shelf life requirement and beyond (4 years) for several MRE retort items (pork sausage, chicken chunks, beef strips in sauce), as indicated by sensory results, with no significant difference in sensory scores for the foil-based retort pouch vs the non-foil Toppan pouch. Barrier data collected as part of this study also showed OTR and WVTR exceeding the requirements and

	fluctuating over the course of the study.
Bibliography Type:	Description: (Last Updated: 06/22/2021)